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Illuminating
ENGINEERING SOCIETY

TECHNICAL MEMORANDUM: LUMINAIRE CLASSIFICATION SYSTEM FOR OUTDOOR LUMINAIRES

AN AMERICAN NATIONAL STANDARD

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Publication of this Technical Memorandum
has been approved by the IES.
Suggestions for revisions
should be directed to the IES.

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The IES Testing Procedures Committee

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Foreword

This Technical Memorandum defines a classification system for outdoor luminaires that provides information to lighting professionals regarding the lumen distribution within solid angles of specific interest. The lumens within these solid angles are intended to be one of the metrics used to evaluate luminaire optical distribution, including the potential for light pollution and obtrusive light, but not as the only metric that should be evaluated. Light pollution and obtrusive light result not only from the optical characteristics of the luminaires, but also from the application of those luminaires within an outdoor site or roadway. A detailed evaluation of the lighting performance for the outdoor site should be based not only on the luminaire optics, but also on overall system design, including luminaire locations, utilization of light where it is needed, lighting quality, visual tasks, aesthetics, safety requirements, and security issues.

1.0 Introduction

Outdoor lighting serves a variety of purposes that include providing light for nighttime visual activities, contributing to safety and security, and enhancing the beauty of architecture, monuments, sculpture, and landscape. Outdoor lighting also serves to improve driving visibility on roadways. Nighttime lighting can enhance social experiences and revitalize the economy of a municipal district. However, a careful selection of lighting equipment is critical to ensure that the positive aspects of outdoor lighting do not simultaneously create a nuisance for local resi-dents. The issues of light pollution, glare, natural habitat, and the nighttime environment are best addressed when meaningful data regarding luminaire optics can be considered as the lighting application is designed.

2.0 Background on Outdoor Luminaire Classifications

Historically, the primary outdoor lighting considerations have related to meeting or exceeding recommended illuminance levels, providing uniform lighting, and minimizing glare. The IES cutoff classification system was redefined in 1963 and published in a revision to the *American National Standard Practice for Roadway Lighting* (now published as *ANSI/IES RP-8-18, Recommended Practice for Design and Maintenance of Roadway and Parking Facility Lighting*[1]) as a method for defining luminaire light distributions. At that time, luminaire light distribution was defined in three ways: 1) the lateral beam width continued to be defined as Types I through V, but the method of determination was redefined; 2) the vertical angle of maximum candlepower (short, medium or long); and 3) to a limited extent, the degree of "glare" control defined by high-angle intensity (cutoff, semi-cutoff, and non-cutoff). The classification "full cutoff" was added in the late 1990s to describe a luminaire with intensity limits meeting the "cutoff" classification, but limiting the optics to only those with no intensity distribution at or above 90 degrees (i.e., no uplight). All four cutoff classifications (full cutoff, cutoff, semi-cutoff, and non-cutoff) were defined and illustrated ANSI/IES RP-8-00 (published in 2000 and reaffirmed in 2005).

The IES cutoff classifications were based only on intensities at or above 80 degrees, rather than on luminaire lumens. The system served a valuable purpose to identify products with high-angle brightness. However, the system had been used for purposes well beyond the technical intent. Full cutoff lighting was often cited as the best system to control light pollution. However, there had been limited consideration related to the ability for full cutoff optics to distribute light at angles necessary to illuminate vertical objects. In addition, the range of performance within each IES cutoff category can result in drastically different percentages of uplight, potentially contributing to sky glow. For example, the analysis of commercial luminaires shown in **Table 2-1** illustrates that within three of the classifications, the luminaire can have very little or a significant amount of uplight. A common misconception is that only luminaires with a higher degree of cutoff (full cutoff) will have minimal uplight, but this is not always the case. Some cutoff, semi-cutoff or even non-cutoff luminaires can have minimal uplight, but do not meet the intensity restrictions at 80 degrees, thus resulting in a lower cutoff classification.

Table 2-1. Upward Light Distribution of Commercial Luminaires by Cutoff Classification

IES Cutoff Classification	Potential Range of Upward Light Distribution (Percentage of Luminaire Lumens)
Full cutoff	0%
Cutoff	0% - 20%
Semi-cutoff	0% - 40%
Non-cutoff	2% - 100%

With increasing concerns among municipalities regarding nuisance light, the IES determined that there was a clear need for a system that would provide more-comprehensive data to evaluate the overall distribution of light from a luminaire.

3.0 Luminaire Classification System (LCS)

The Luminaire Classification System (LCS) defines the distribution of light from a luminaire within three primary solid angles. These are further divided into 10 secondary solid angles. LCS can be described as either percent bare lamp lumens or luminaire lumens for each primary and secondary solid angle. It is based in part on IES-funded research.[2] The LCS quantifies light distribution in front of the luminaire, behind the luminaire, and above the luminaire. The system offers the following benefits:

- LCS defines the standard solid angles for evaluation and comparison of outdoor luminaires. It does not provide quantitative lumen limits within each solid angle. It does provide the basic model from which limits for lumens within the solid angles by lighting zone and application type will be defined.
- LCS utilizes existing photometric test data and can be easily reported by manufacturers or incorporated into software tools.
- LCS enables designers to evaluate and compare the distribution of lumens for various types of luminaire optics, thus assisting in the selection of the luminaire most appropriate for the application.

As illustrated in **Figure 3-1**, the primary solid angles defined by the LCS are:

- Forward Light
- Back Light
- Uplight

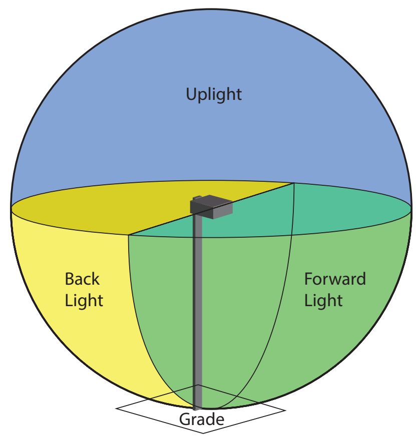


Figure 3-1. The three primary solid angles of the Luminaire Classification System (LCS). (© Illuminating Engineering Society)

The sum of percentages of lamp lumens within these three primary solid angles is equal to the photometric luminaire efficiency.

$$\textit{Photometric Luminaire Efficiency} = \frac{\textit{Forward Light} + \textit{Back Light} + \textit{Uplight}}{\textit{Total Bare Lamp Output}},$$

where all values on the right-hand side are in lumens.

$$\textit{Photometric Luminaire Efficiency}\ (\%) = \textit{Forward Light}\ (\%) + \textit{Back Light}\ (\%) + \textit{Uplight}\ (\%)$$

$$\textit{Trapped Light}\ (\%) = 100\% - \textit{Photometric Luminaire Efficiency}\ (\%)$$

3.1 Intended Use

The LCS metrics are indicators of optical distribution and are intended to be used in conjunction with the IES distribution classifications (Types I, II, III, IV, and V; and Short, Medium, and Long lateral distributions) for a more complete analysis of where the light is distributed.

As previously noted, the LCS is designed to describe the lumen distribution of an individual luminaire. It also provides a convenient method to compare the utilization of available lamp lumens in the three LCS solid angles among similar non-aimable outdoor luminaires.

The lumens within each LCS solid angle provide data that can relate to an evaluation of light trespass and sky glow. However, these issues relate also to the optical distribution of light as a function of the installed characteristics, including location of the luminaires with respect to the property line, installed height, spacing and uniformity of light, and reflective characteristics of the ground materials that may contribute to light reflected into the sky.

The previous IES cutoff classifications (full cutoff, cutoff, semi-cutoff, and non-cutoff) are superseded by the Luminaire Classification System (LCS).

3.2 Solid Angle References

The LCS is based on IES photometric testing procedures. Using these procedures, a web of intensity values is measured around a luminaire, creating a sphere of data points (see **Figure 3-2**). Luminaire lumens are calculated based on the measured intensities in specific solid angles. The term *nadir* refers to the point directly below the luminaire. This IES publication refers to LCS solid angles based on vertical angles referenced from nadir and lateral angles referenced in a counterclockwise direction.

3.3 Forward Light

Forward light describes the lumen distribution in front of the luminaire. The forward light solid angle is defined between 0 and 90 degrees vertical, and from 270 to 90 degrees horizontal (in front of the luminaire). The forward light solid angle is further refined into four vertical secondary solid angles to evaluate the distribution of light

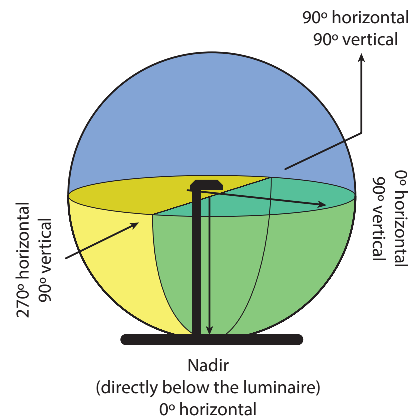


Figure 3-2. Solid angle references are based on a sphere of data points around a luminaire. (© Illuminating Engineering Society)

in front of the luminaire. The forward light secondary solid angles (see **Figure 3-3**) are defined as:

- *Forward light low secondary solid angle (FL):* Percent lamp lumens between 0 and 30 degrees vertical (or luminaire lumens within that solid angle) in front of the luminaire. This is the light emitted from directly below the luminaire that will reach up to 0.6 mounting heights away from the luminaire.
- *Forward light mid secondary solid angle (FM):* Percent lamp lumens between 30 and 60 degrees vertical (or luminaire lumens within that solid angle) in front of the luminaire. This is the light that will reach from 0.6 to 1.7 mounting heights away from the luminaire.
- *Forward light high secondary solid angle (FH):* Percent lamp lumens between 60 and 80 degrees vertical (or luminaire lumens within that solid angle) in front of the luminaire. This is the light that will reach from 1.7 to 5.7 mounting heights away from the luminaire.
- *Forward light very high secondary solid angle (FVH):* Percent lamp lumens between 80 and 90 degrees vertical (or luminaire lumens within that solid angle) in front of the luminaire. This is the light that will reach beyond 5.7 mounting heights away from the luminaire.

3.4 Back Light

Back light describes the lumen distribution behind the luminaire. The back light solid angle is defined between 0 and 90 degrees vertical, and 90 to 270 degrees

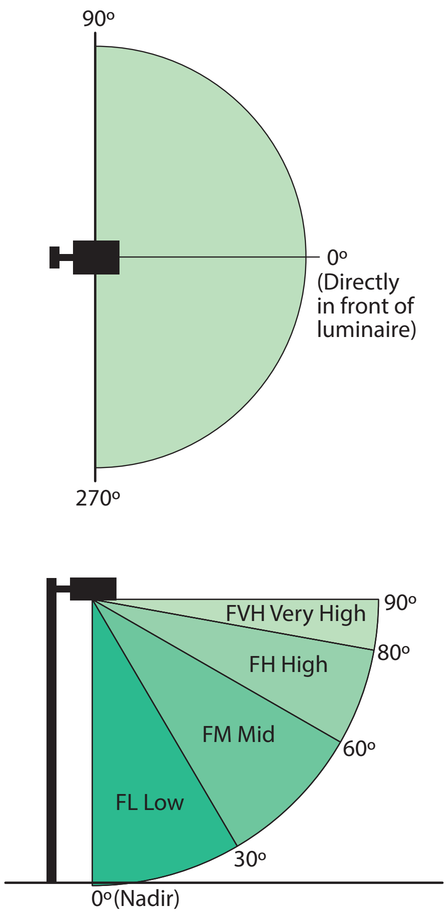


Figure 3-3. Top: plan view for forward solid angle; bottom: section view for forward solid angle.

horizontal (behind the luminaire). This solid angle can be used to evaluate light trespass when luminaires are located near the property line. When luminaires are located interior to a site, a luminaire's back light may or may not be considered offensive light. The back light solid angle is further refined into four vertical secondary solid angles to evaluate the distribution of light behind the luminaire. The back light secondary solid angles (see **Figure 3-4**) are defined as:

- *Back light low secondary solid angle (BL):* Percent lamp lumens between 0 and 30 degrees vertical (or luminaire lumens within that solid angle) behind the luminaire. This is the light emitted from directly below the luminaire that will reach up to 0.6 mounting heights away from the luminaire.
- *Back light mid secondary solid angle (BM):* Percent lamp lumens between 30 and 60 degrees vertical (or luminaire lumens within that solid angle) behind the luminaire. This is the light that will reach from 0.6 to 1.7 mounting heights away from the luminaire.
- *Back light high secondary solid angle (BH):* Percent lamp lumens between 60 and 80 degrees vertical (or luminaire lumens within that solid angle) behind the luminaire. This is the light that will reach from 1.7 to 5.7 mounting heights away from the luminaire.
- *Back light very high secondary solid angle (BVH):* Percent lamp lumens between 80 and 90 degrees vertical (or luminaire lumens within that solid angle) behind the luminaire. This is the light that will reach beyond 5.7 mounting heights away from the luminaire.

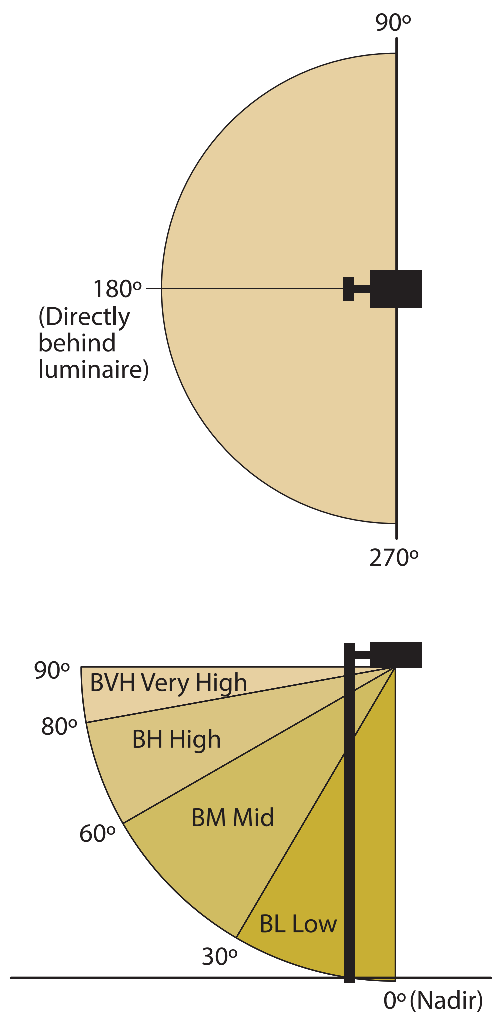


Figure 3-4. Top: plan view for back light solid angle; bottom: section view for back light solid angle.

3.5 Uplight

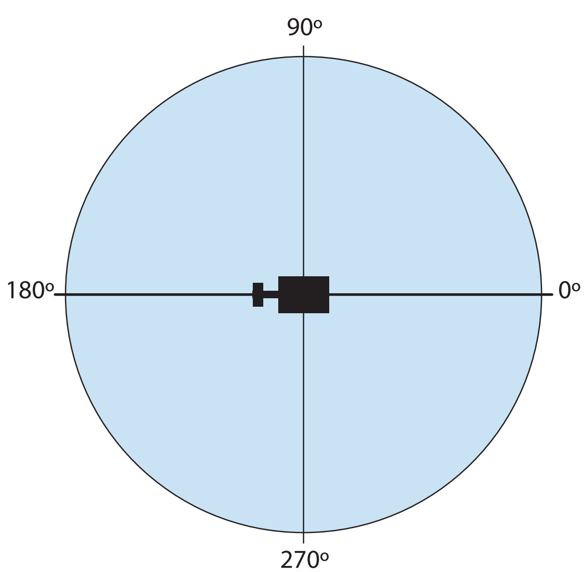
Uplight describes the lumen distribution above the luminaire. The uplight solid angle is defined between 90 and 180 degrees vertical, and 0 to 360 degrees horizontal, around the entire luminaire. Uplight is a major contributor to sky glow. The overall impact on sky glow is a function of the angle of light above the horizontal, atmospheric scattering of the light, and geographic location.[3] The uplight solid angle does not account for the directional impact on sky glow, nor does it quantify the impact from light reflected from ground surfaces and adjacent structures.

The uplight solid angle is further refined into two vertical secondary solid angles to evaluate the distribution of light at or near horizontal and that directly above the luminaire. The uplight secondary solid angles (see **Figure 3-5**) are defined as:

- *Uplight low secondary solid angle (UL):* Percent lamp lumens between 90 and 100 degrees vertical (or luminaire lumens within that solid angle), 360 degrees around the luminaire. Light emitted at or slightly above 90 degrees will impact the sky glow as observed far from a city.[3]
- *Uplight high secondary solid angle (UH):* Percent lamp lumens between 100 and 180 degrees vertical (or luminaire lumens within that solid angle), 360 degrees around the luminaire. Light emitted at angles above 100 degrees will impact sky glow directly over the city.[3]

4.0 Application Examples of the Luminaire Classification System

This section provides examples of how the LCS can be used to compare the solid-angle lumen distributions of luminaires. When evaluating luminaires, the LCS solid angles that are of specific interest for the requirements of the application should be considered. When luminaires are located interior to a site, the evaluation of a luminaire distribution may or may not consider the back light as offensive light. The tables in the following subsections provide samples of values for illustrative purposes only. They do not suggest specific product types for particular applications, nor do they imply IES endorsement of any manufacturer.



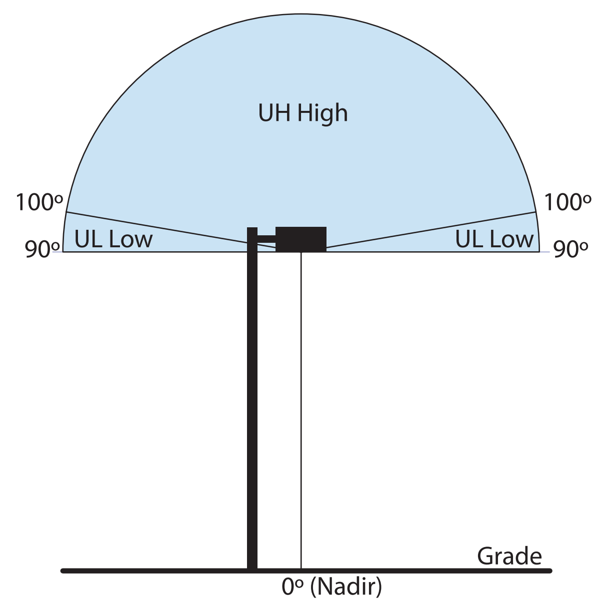


Figure 3-5. Left: plan view for uplight solid angle; right: section view for uplight solid angle. (© Illuminating Engineering Society)

4.1 "Shoebox" Type Luminaires

Table 4-1 provides optical evaluation examples of a few general area "shoebox" luminaires for parking lots that would be located at or near the property line. In comparing them, key items for consideration are:

- There is an increase in very high-angle lumens with products that use a drop lens or sag lens.
- Luminaires that include a house-side shield or sharp cutoff reduce the amount of back light that would contribute to light trespass.
- Increased back light optical control increases the amount of trapped light, resulting in a lower photometric efficiency.
- All of the products in this example have little to no uplight.
- Several non-optical attributes should also be considered in the overall selection of the luminaire (including but not limited to uniformity of light, ability to light vertical objects, color, energy, and cost).

Table 4-1. Optical Evaluation Examples of General Area "Shoebox" Luminaires

400 W MH Type IV					
	Standard optics	House side shield	Sharp cutoff optics	Standard Optics	House side shield
Forward Light					
Luminaire Lumens	13,994	12,747	14,802	13,647	13,179
% Lamp Lumens	43.7%	39.8%	46.3%	37.9%	36.6%
FL (0°-30°)	4.7%	4.4%	4.7%	2.8%	2.2%
FM (30°-60°)	17.8%	14.8%	23.7%	12.5%	11.2%
FH (60°-80°)	21.0%	19.8%	17.6%	21.4%	21.8%
FVH (80°-90°)	0.2%	0.8%	0.2%	1.3%	1.4%
Back Light					
Luminaire Lumens	8,282	3,953	1,896	13,647	1,786
% Lamp Lumens	25.9%	12.4%	5.9%	37.9%	5.0%
BL (0°-30°)	4.6%	3.5%	2.2%	2.8%	0.9%
BM (30°-60°)	14.6%	6.6%	3.2%	12.5%	1.7%
BH (60°-80°)	6.4%	2.2%	0.4%	21.4%	2.2%
BVH (80°-90°)	0.2%	0.1%	0.0%	1.3%	0.1%
Uplight					
Luminaire Lumens	0	0	0	87	88
% Lamp Lumens	0.0%	0.0%	0.0%	0.2%	0.2%
UL (90°-100°)	0.0%	0.0%	0.0%	0.2%	0.2%
UH (100°-180°)	0.0%	0.0%	0.0%	0.0%	0.0%
Trapped Light					
Luminaire Lumens	9,724	15,300	15,301	8,620	20,946
% Lamp Lumens	30.4%	47.8%	47.8%	23.9%	58.2%

Table note: Trapped light is the difference between one hundred percent and the total of the forward light, back light, high angle light, and uplight. It represents the amount of light that is not emitted from the luminaire. A high percentage of trapped light is indicative of an optical system with lower efficiency.

4.2 Decorative Street Lighting

Table 4-2 provides optical evaluation examples of a few decorative streetlights. In comparing them, key items of consideration are:

- The flat-lens, opaque-sided luminaire has no uplight.
- The streetlight with no optical control has a significant amount of very high-angle light.
- Back light in this example is considered useful light because the luminaires are not placed near a property line.
- Increased optical control increases the amount of trapped light, resulting in a lower photometric efficiency.
- The amount of uplight varies significantly but may be reduced with optical control (although the teardrop luminaire would appear to potentially have significant uplight, it has less than one percent).
- Several non-optical attributes should also be considered in the overall selection of the luminaire (including but not limited to aesthetics, ability to light vertical objects, color, energy, and cost).

Table 4-2. Optical Evaluation Examples of Decorative Street Lights

175 W MH				
	Fully shielded (Type III)	Glass tear drop (Type V)	Glass acorn internal optics (Type V)	Acrylic acorn (Type V)
Forward Light				
Luminaire Lumens	5,325	4,521	3,459	3,551
% Lamp Lumens	41.6%	32.3%	25.4%	26.1%
FL (0°-30°)	4.8%	3.8%	0.4%	0.5%
FM (30°-60°)	19.1%	14.9%	7.0%	4.0%
FH (60°-80°)	17.5%	12.9%	15.8%	13.2%
FVH (80°-90°)	0.2%	0.7%	2.2%	8.4%
Back Light				
Luminaire Lumens	2,837	4,521	3,459	3,551
% Lamp Lumens	22.2%	32.3%	25.4%	26.1%
BL (0°-30°)	4.4%	3.8%	0.4%	0.5%
BM (30°-60°)	13.8%	14.9%	7.0%	4.0%
BH (60°-80°)	3.8%	12.9%	15.8%	13.2%
BVH (80°-90°)	0.2%	0.7%	2.2%	8.4%
Uplight				
Luminaire Lumens	0	33	1,134	4,328
% Lamp Lumens	0.0%	0.2%	8.3%	31.8%
UL (90°-100°)	0.0%	0.1%	2.2%	8.9%
UH (100°-180°)	0.0%	0.1%	6.1%	23.0%
Trapped Light				
Luminaire Lumens	4,638	4,924	5,549	2,170
% Lamp Lumens	36.2%	35.2%	40.8%	16.0%

4.3 Roadway Luminaires

Table 4-3 provides optical evaluation examples of a few roadway luminaires. In comparing them, key items of consideration are:

- All the luminaires distribute a similar percentage of light in the forward direction.
- All the luminaires have about the same amount of back light.
- All the luminaires have about the same amount of very high-angle light.
- The drop-lens cobra head has slightly more uplight.
- The drop-lens cobra head has less trapped light.
- Several non-optical attributes should also be considered in the overall selection of the luminaire (including but not limited to the utilization of light onto the roadway, luminance, the overall distribution and uniformity of light, color, energy, and cost).

This particular example is a case where the products appear to have very similar optical characteristics; however, a more detailed evaluation may show differences in the distribution of light on the roadway and glare characteristics.

Table 4-3. Optical Evaluation Examples of Roadway Luminaires

HPS Type II	 100 Watt	 150 Watt	 150 Watt
Forward Light			
Luminaire Lumens	5,365	8,589	9,188
% Lamp Lumens	56.5%	53.7%	57.4%
FL (0°-30°)	6.8%	5.0%	11.5%
FM (30°-60°)	24.5%	27.3%	27.6%
FH (60°-80°)	24.8%	20.6%	17.1%
FVH (80°-90°)	0.4%	0.8%	1.3%
Back Light			
Luminaire Lumens	1,985	3,447	3,832
% Lamp Lumens	20.9%	21.5%	24.0%
BL (0°-30°)	4.2%	4.1%	5.2%
BM (30°-60°)	10.7%	12.3%	11.8%
BH (60°-80°)	5.7%	4.4%	6.0%
BVH (80°-90°)	0.3%	0.8%	0.9%
Uplight			
Luminaire Lumens	42	0	390
% Lamp Lumens	0.4%	0.0%	2.4%
UL (90°-100°)	0.2%	0.0%	1.1%
UH (100°-180°)	0.2%	0.0%	1.3%
Trapped Light			
Luminaire Lumens	2,108	3,964	2,589
% Lamp Lumens	22.2%	24.8%	16.2%

5.0 Summary

The LCS provides a useful tool to evaluate the overall distribution of lumens into key solid angles. However, as illustrated in these examples, the LCS should not be the sole procedure used in evaluating lighting quality. There are other considerations that should be evaluated, including (but not limited to) illuminance levels, uniformity, luminance, glare, aesthetics, color, energy efficiency, cost, safety, security, distribution of light onto areas where it is needed, and specifics appropriate for the intended application.

Annex A – Back Light, Uplight, and Glare (BUG) Ratings

This Annex is not part of ANSI/IES TM-15-20, but is included for informational purposes only.

The following Back Light, Uplight, and Glare ratings may be used to evaluate luminaire optical performance related to light trespass, sky glow, and high-angle brightness control. These ratings are based on a zonal lumen calculations for secondary solid angles defined in TM-15. The zonal lumen thresholds listed in **Tables A-1**, **A-2**, and **A-3** are based on data from photometric testing procedures approved by the Illuminating Engineering Society for outdoor luminaires (ANSI/IES LM-31-20[4] or ANSI/IES LM-35-20[5]).

Table A-1. Backlight Ratings (maximum zonal lumens)

		Backlight Rating					
	Secondary Solid Angle	B0	B1	B2	B3	B4	B5
Backlight / Trespass	BH	110	500	1000	2500	5000	>5000
	BM	220	1000	2500	5000	8500	>8500
	BL	110	500	1000	2500	5000	>5000

Table A-2. Uplight Ratings (maximum zonal lumens)

		Uplight Rating					
	Secondary Solid Angle	U0	U1	U2	U3	U4	U5
Uplight / Skyglow	UH	0	10	50	500	1000	>1000
	UL	0	10	50	500	1000	>1000

Table A-3. Glare Ratings (maximum zonal lumens)

Glare Rating for
Asymmetrical Luminaire Types (Type I, Type II, Type III, Type IV)

	Secondary Solid Angle	G0	G1	G2	G3	G4	G5
Glare / Offensive Light	FVH	10	100	225	500	750	>750
	BVH	10	100	225	500	750	>750
	FH	660	1800	5000	7500	12000	>12000
	BH	110	500	1000	2500	5000	>5000

Glare Rating for
Quadrilateral Symmetrical Luminaire Types (Type V, Type V Square)

	Secondary Solid Angle	G0	G1	G2	G3	G4	G5
Glare / Offensive Light	FVH	10	100	225	500	750	>750
	BVH	10	100	225	500	750	>750
	FH	660	1800	5000	7500	12000	>12000
	BH	660	1800	5000	7500	12000	>12000

Notes for Tables A-1, A-2, and A-3:

1. Any one rating is determined by the maximum rating obtained for that table. For example, if the BH zone is rated B1, the BM zone is rated B2, and the BL zone is rated B1, then the *Back Light rating for the luminaire is B2*.
2. To determine BUG ratings, the photometric test data shall include data in the upper hemisphere unless no light is emitted above 90 degrees vertical (for example, if the luminaire has a flat lens and opaque sides), per the IES Testing Procedures Committee recommendations.
3. It is recommended that the photometric test density include values at least every 2.5 degrees vertically. If a photometric test does not include data points every 2.5 degrees vertically, the BUG ratings shall be determined based on appropriate interpolation.
4. A "quadrilaterally symmetric" luminaire shall meet one of the following definitions:

   a. A Type V luminaire is one with a distribution that has circular symmetry, defined by the IES as being essentially the same at all lateral angles around the luminaire.

   b. A Type VS luminaire is one where the zonal lumens for each of the eight horizontal octants (0 – 45, 45 – 90, 90 – 135, 135 – 180, 180 – 225, 225 – 270, 270 – 315, 315 – 360) are within ±10 percent of the average zonal lumens of all octants.

"BUG" Rating Example:
A 250-watt metal halide area luminaire, Type IV forward-throw optical distribution.

Based on the photometric test data, the luminaire has this zonal lumen distribution:

Forward Light	Lumens	% Lamp Lumens
FL (0 - 30 degrees)	1618	5.9%
FM (30 - 60 degrees)	6093	22.2%
FH (60 – 80 degrees)	3748	13.6%
FVH (80 – 90 degrees)	27	0.1%
Back Light		
BL (0 – 30 degrees)	985	3.6%
BM (30 – 60 degrees)	930	3.4%
BH (60 – 80 degrees)	136	0.5%
BVH (80 – 90 degrees)	16	0.1%
Uplight		
UL (90 – 100 degrees)	0	0.0%
UH (100 – 180 degrees)	0	0.0%

Backlight rating: Determine the highest rating where the lumens for each the secondary solid angles do not exceed the threshold lumens from **Table A-1**. In this example, the backlight rating would be **B2** based on the BL lumen limit.

Uplight rating: Determine the highest rating where the lumens for each of the secondary solid angles do not exceed the threshold lumens from **Table A-2**. In this example, the uplight rating would be **U0** based on the FVH and BVH limits.

Glare rating: Determine the highest rating where the lumens for each of the secondary solid angles do not exceed the threshold lumens from **Table A-3** for a Type IV distribution. In this example, the glare rating would be **G2** based on the FH upper limit.

Therefore, the BUG rating for this luminaire would be **B2 U0 G2**.

REFERENCES

1 Illuminating Engineering Society. ANSI/IES RP-8-18, Recommended Practice for Design and Maintenance of Roadway and Parking Facility Lighting. New York: IES; 2018.

2 McColgan MW, Bullough JD, Van Derlofske J, Rea MS. LESS: Luminaire Evaluation and Selection System. Troy, NY: Lighting Research Center, Rensselaer Polytechnic Institute; 2005 Jul.

3 Cinzano P, Diaz FJ. The artificial sky luminance and the emission angles of the upward light flux. J Italian Astro Soc., 2000;71(1):251. Online: http://www.lightpollution.it/cinzano/en/index.html. (Accessed 2020 Apr 15). Click on "Measuring and modelling light pollution" in the description of the second document listed (spelling is as noted). This will bring up an index containing several clickable article titles.

4 Illuminating Engineering Society. ANSI/IES LM-31-20, Approved Method: Photometric Testing for Roadway and Area Lighting Luminaires Using Incandescent Filament or High Intensity Discharge Lamps. New York: IES; 2020.

5 Illuminating Engineering Society. ANSI/IES LM-35-20, Approved Method: Photometric Testing of Floodlights Using High Intensity Discharge or Incandescent Filament Lamps. New York: IES; 2020.

Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: ______________________________

   Affiliation: ______________________________

   Address: ______________________________

   City: ______________ State: ________ Zip: ________ Country: ______________

   Telephone: ______________________________

   Fax: ______________________________

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I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

Submitter's signature: ______________________________ Date: ______________

2. Title of publications and year published ______________________________
3. Clause (section), sub-clause or paragraph number; and page number: ______________________________
4. My proposal (check one):
   [ ] Change to read as follows
   [ ] Delete and substitute as follows
   [ ] Add new text as follows
   [ ] Delete without substitution

Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

____

____

____

6. Reason and substantiation:

____

____

____

Select as applicable:

[ ] Additional pages are attached. Number of additional pages:

[ ] Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:

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